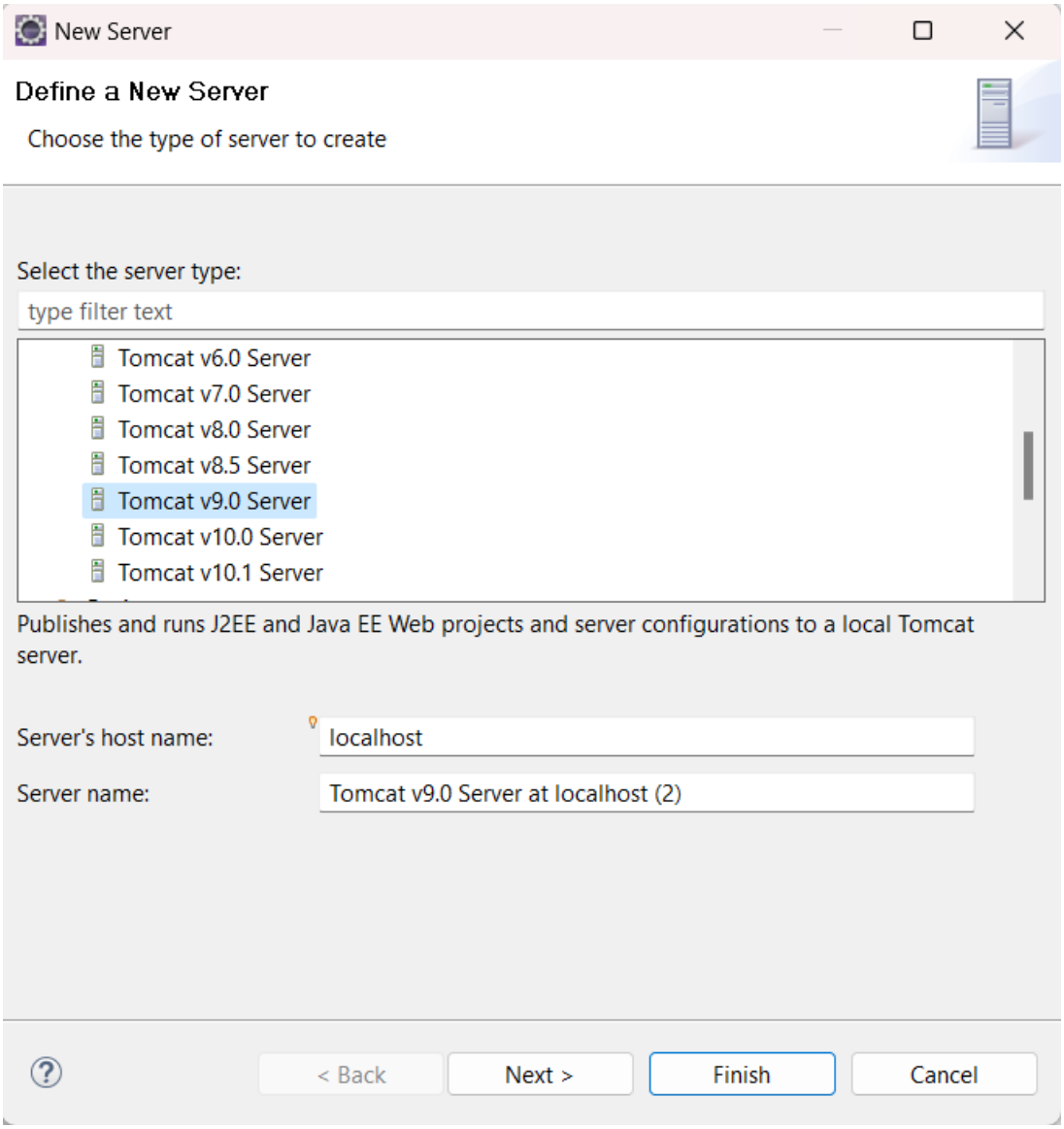
WS deployment with Eclipse and Tomcat

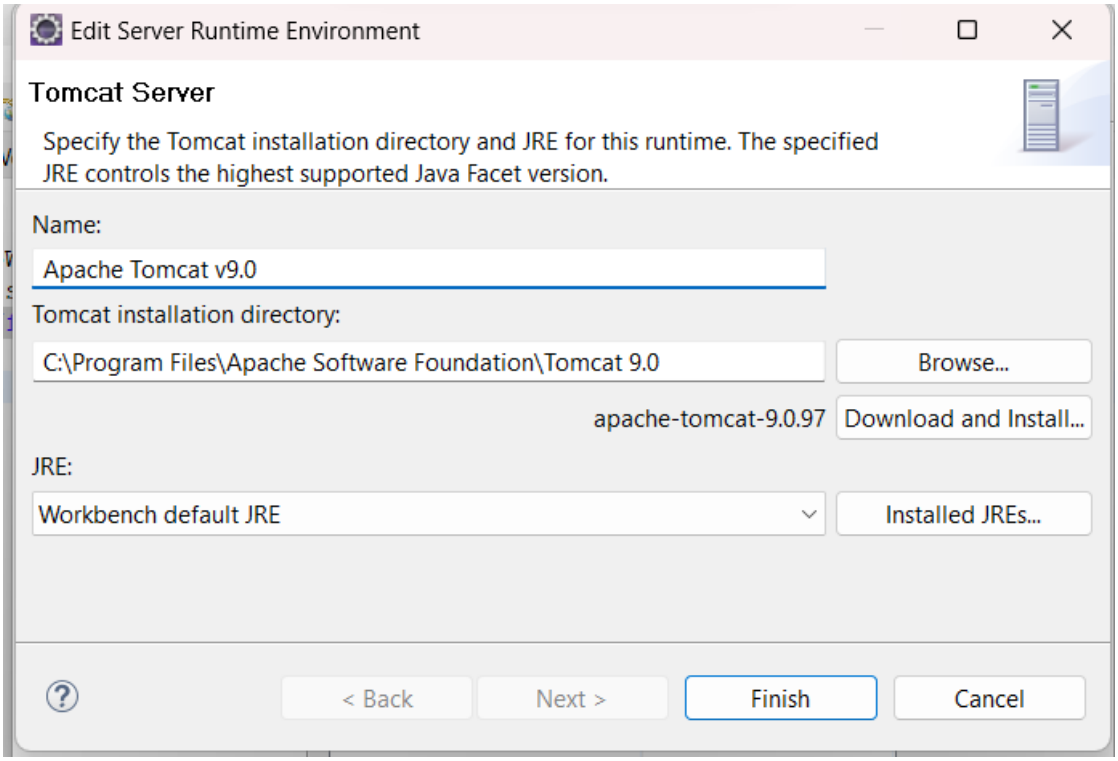
This tutorial shows you how to create a web service using Eclipse and deploy it on the Tomcat server.

Setting up Tomcat on Eclipse

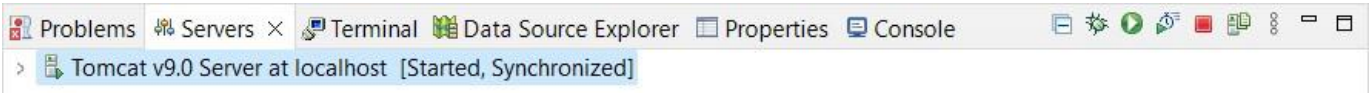
1. Download and install Tomcat (version 9.0.93) on your machine.
2. Open the server view on Eclipse by the menu **Windows→Show View→Other→Servers**.
3. Right click on the Server view select **New→Server**.
   Let the Server's host name **localhost**,
   select the Server type **Apache→Tomcat v9.0 Server**.
   Click **Next**.



4. Specify the installation directory, path to\apache-tomcat-9.0.93.
   Click **Finish**.



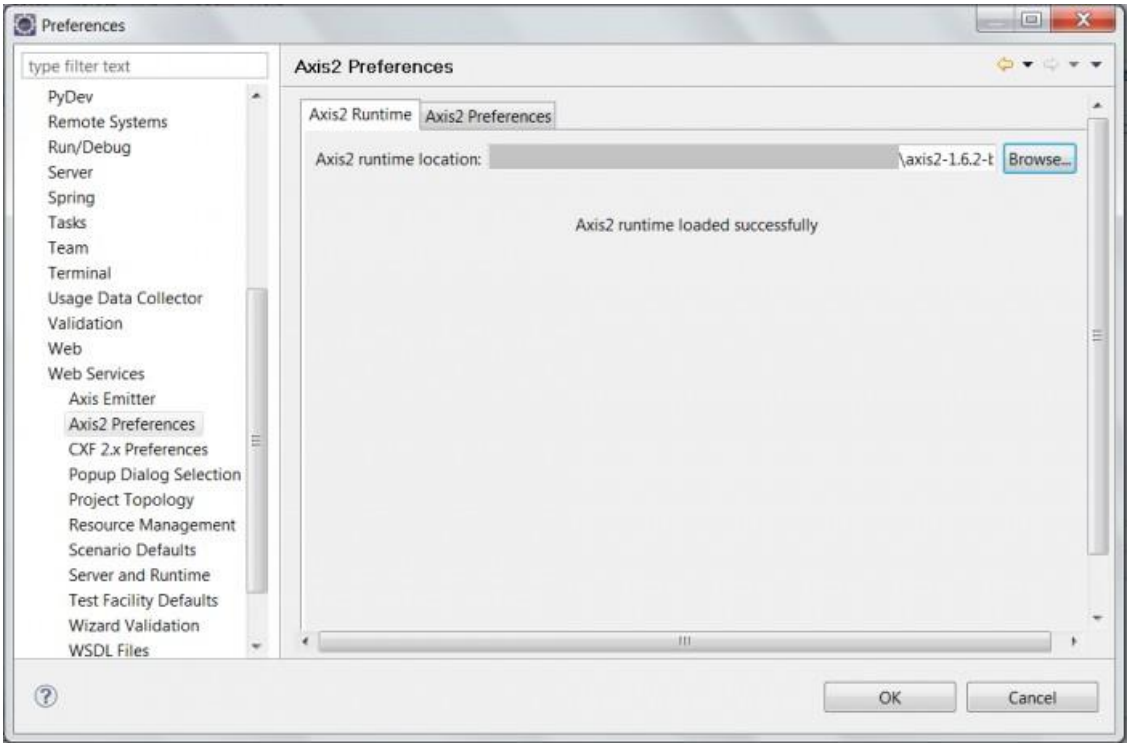
5. You can test the server by clicking on the **Start** and **Stop** button at the **Server** view.
   You can also reconfigure the server by selecting **Windows→Preferences→Server→Runtime Environments**.



Setting up Axis2 on Eclipse

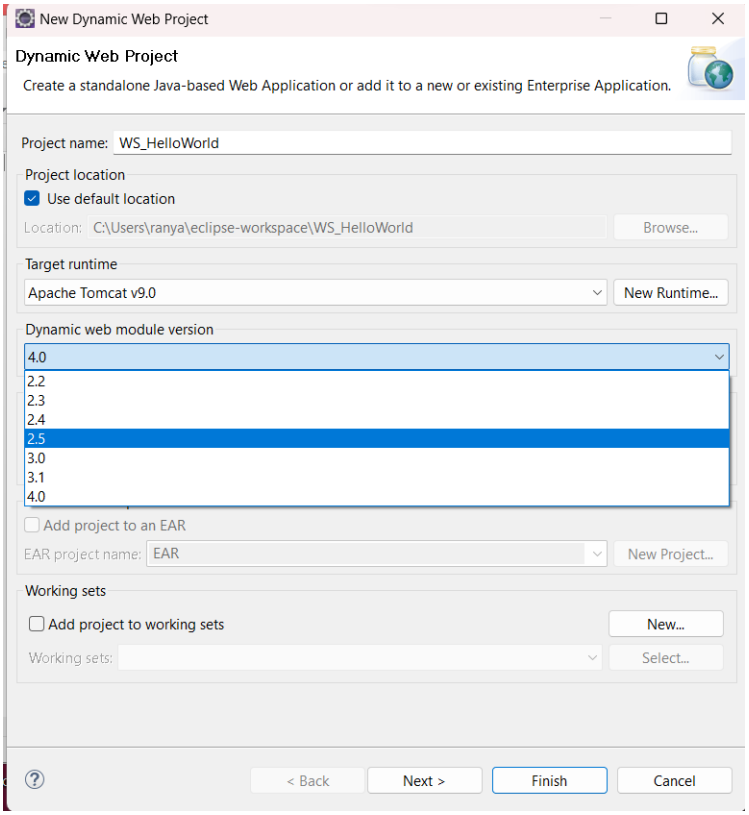
Apache Axis2 is Web services/SOAP/WSDL engine and is the successor of Apache Axis. It supports both SOAP 1.1 and 1.2.

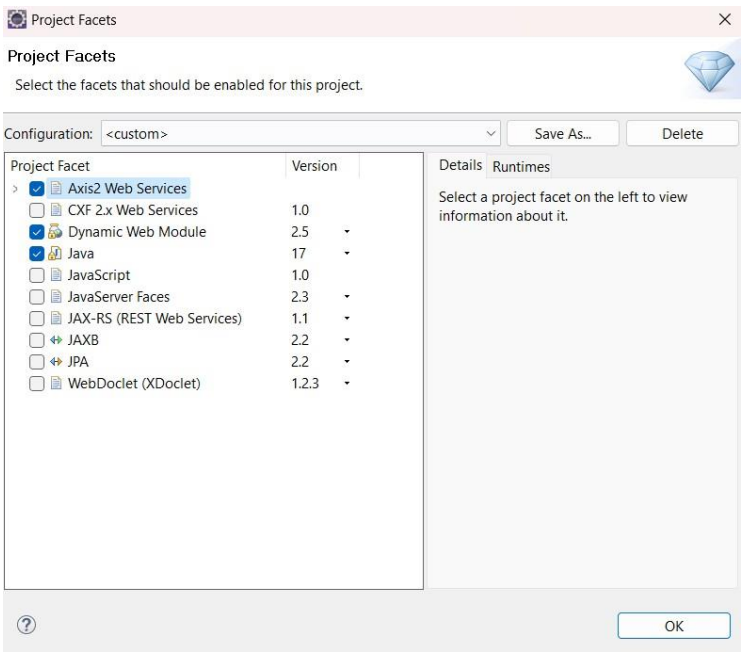
1. Download and extract Axis2 1.6.2 binary distribution and extract it on your machine.
2. Download the axis.war file the add it under path to**\Tomcat 9.0\webapps**
3. Select **Window→Preferences** on eclipse.
   Go to **Web services→Axis2 Preferences**.
   In **Axis2 runtime location** browse to your axis2 home folder (path to/ axis2-1.6.2).
   Click **Ok**.



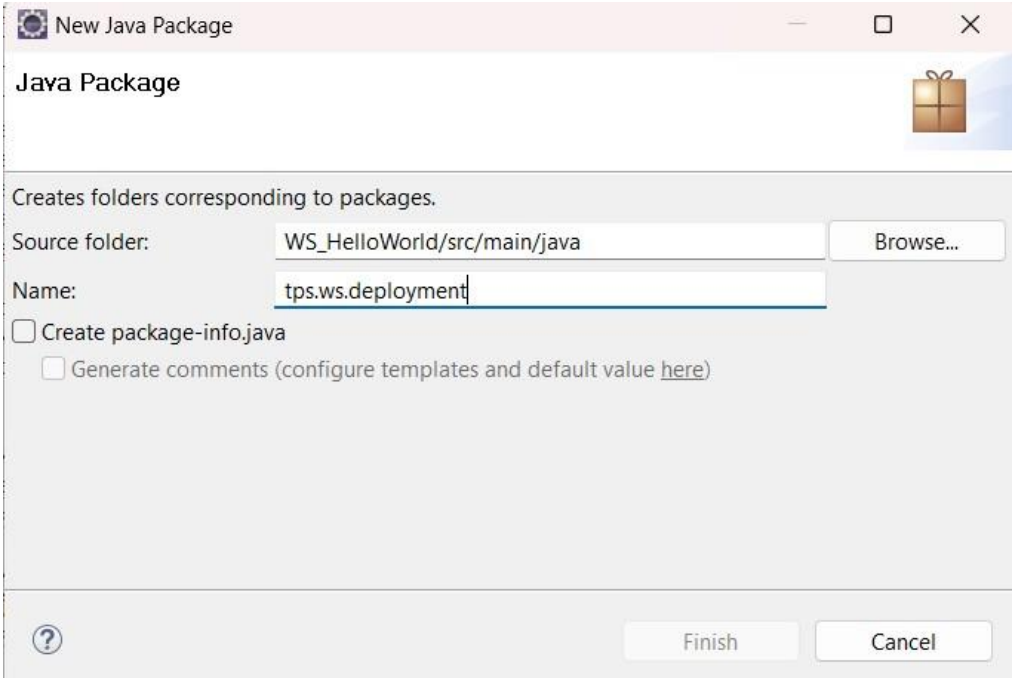
HelloWorld service deployment

1. Create a Dynamic Web project named **WS_HelloWorld** by selecting **File→New→Dynamic Web Project**.
   Keep the Target Runtime as Apache Tomcat **v9.0**.
   Select the **Dynamic web module version** as **2.5**.
   In the **Configuration** part, click on **modify** and check **Axis2 Web Services** then click **Ok**.
   Click **Finish**.

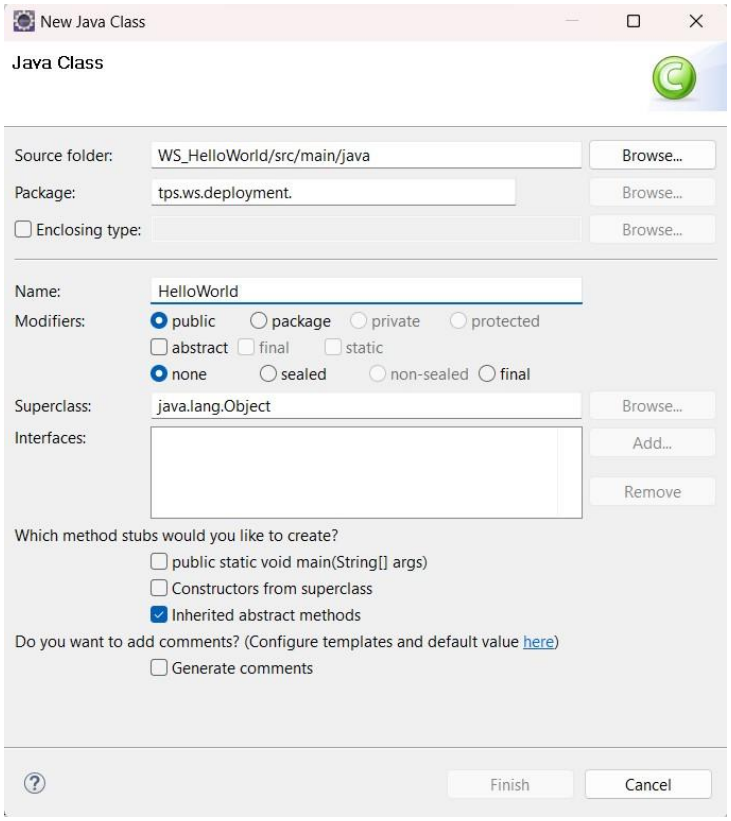




2. Open the **WS_HelloWorld** project, right click on the **Java Resources** directory, select **New→Package**.
   Name it as **tps.ws.deployment**.
   Click **Finish**.



3. Create a Java class named **HelloWorld** under the package **tps.ws.deployment**.

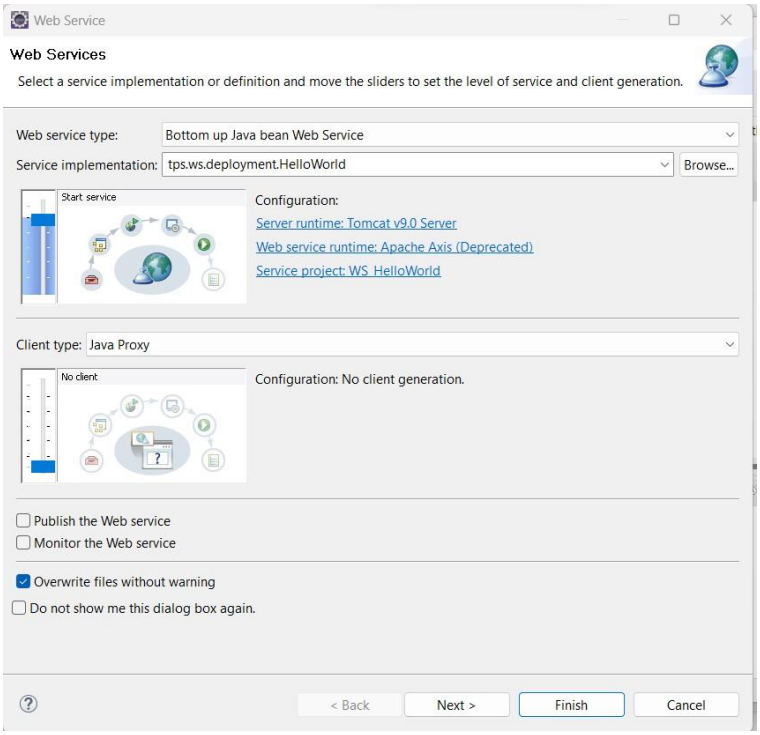


4. In the HelloWorld class, create the function **sayHello** which returns the string **"Hello Word"**.

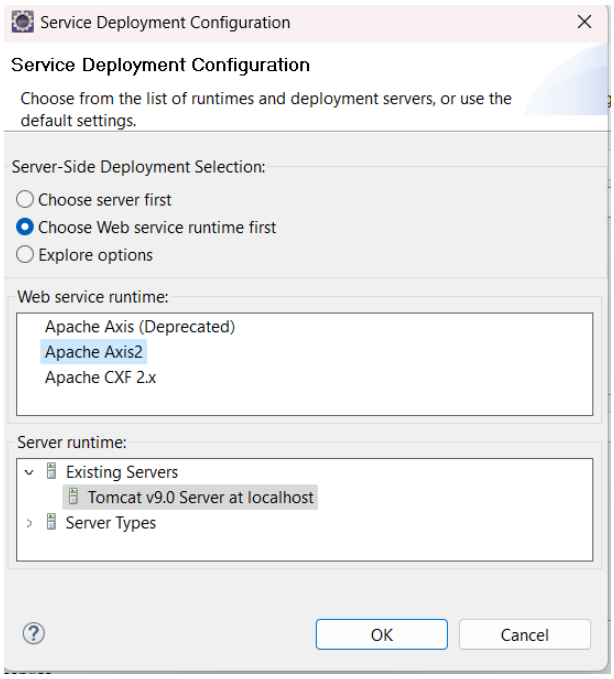
```
package tps.ws.deployment;
    public class HelloWorld {
        public String sayHello()
```

{ return("Hello World ") ;} }

5. Right click on the HelloWorld class, select **Web Sevices→Create Web service**.

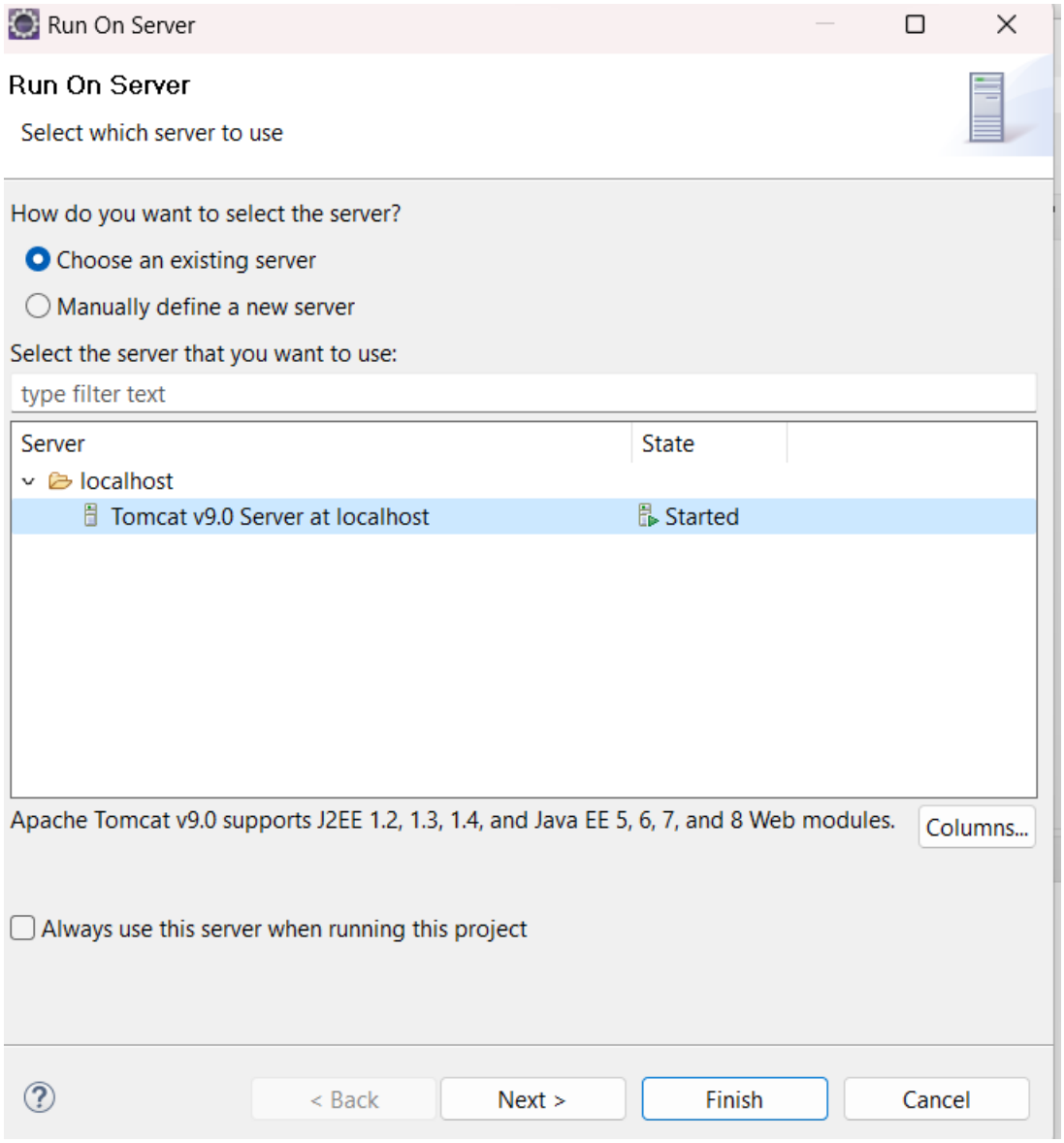


6. In the **Configuration** part, select **Web service runtime: Apache Axis** and select **Apache Axis2** in the **Web service runtime** tab, then click **Ok**.
   Click **Finish**.

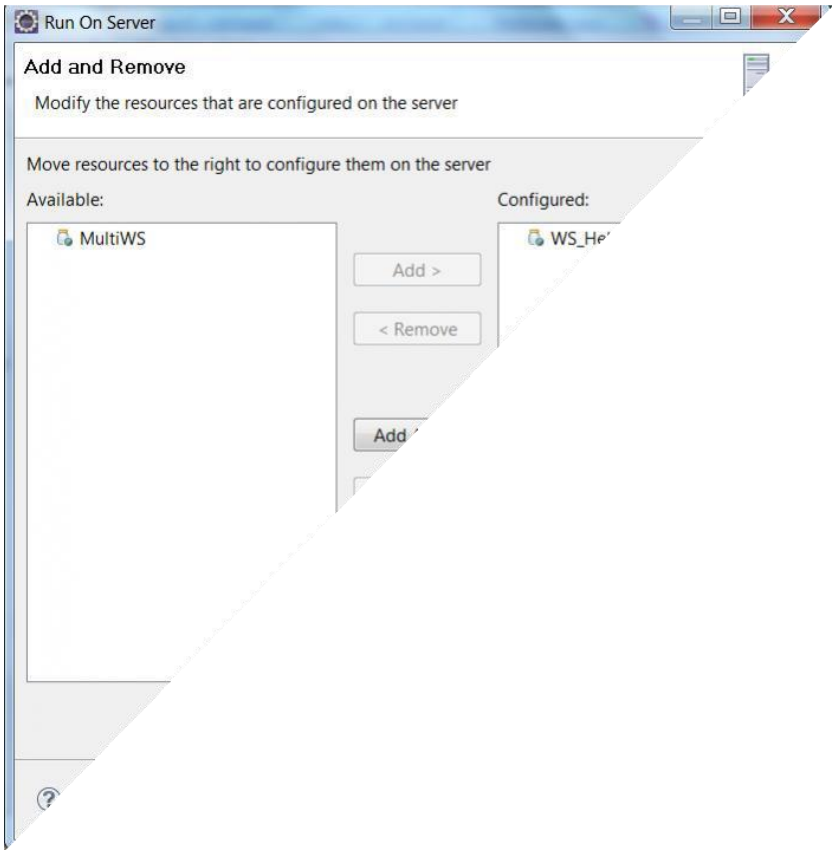


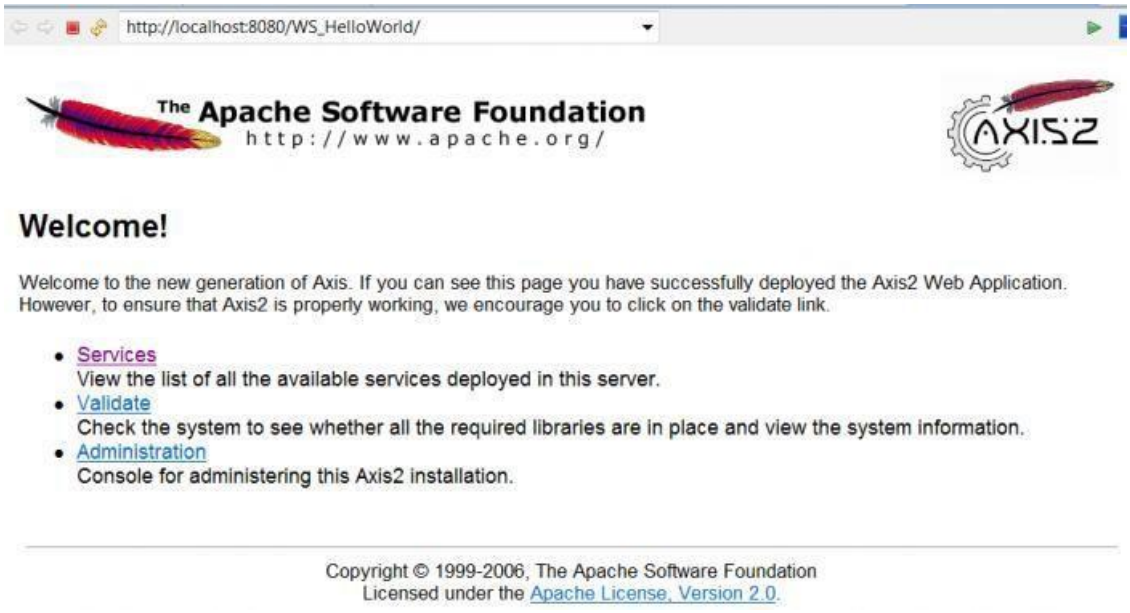
➔ The WS skeleton is automatically generated, the service is automatically published on the Eclipse's Tomcat server, and the Tomcat server automatically run.

7. The WSDL file in axis2 is dynamically generated on service deployment.
   To access it from eclipse Web service explorer, right click on **WS_HelloWorld** project, then **Run as→Run on a server**.
   Choose **Tomcat v9.0**.
   Click **Next**.



8. Make sure that **WS_HelloWorld** project is in the **Configured** list.
   If not **add** it from **Available** list to **Configured** list.
   Click **Finish**.



9. A **Welcome** page appears.
10. 

11. Select **services**. A list of available services is presented. Click on **HelloWorld** service to see the corresponding **WSDL** file.

12. To invoke the **sayHello** operation, enter the following url in eclipse web service explorer **http://localhost:8080/WS_HelloWorld/services/HelloWorld/sayHello**.